

GC University Lahore

BS Computer Science

Examination: Final 2022

Semester: 7th

COURSE CODE: CS-3224

TITLE: Natural Language Processing

Time allowed: 155 minutes

Marks: 40

Name: _____

Roll No _____

Subjective

Attempt any Four Questions

Q. No.	Questions	Marks						
2	Calculate the maximum entropy model for email spam detection Example: Email as input x, d as two classes spam and not spam Features <table><thead><tr><th>Feature</th><th>Present</th><th>Weight</th></tr></thead><tbody><tr><td>(Spam)</td><td>Weight</td><td>(Not Spam)</td></tr></tbody></table> F1: Look at who the email is addressed to F2: check the "from" email address F3: check the greetings F4: Look at the subject of the email F5: email content are money, lottery, win, prize F6: Bad grammar and spelling F7: No phone number in the signature	Feature	Present	Weight	(Spam)	Weight	(Not Spam)	10
Feature	Present	Weight						
(Spam)	Weight	(Not Spam)						
3	Discuss the knowledge based, supervised and unsupervised techniques for word sense disambiguation?	10						
4	a) Briefly discuss named entity recognition with the help of examples? b) Briefly discuss the difference between Information Extraction & Information Retrieval approaches?	5 + 5						
5	a) Write down the basic issues of machine translation? b) What is the role of Treebanks in NLP?	5 + 5						
6	Consider the following grammar: Show three parse trees that would be derived for the sentence Ahmad flew to Karachi and Lahore in February.	10						

S -> NP VP

VP -> Verb NP

VP -> Verb PP

VP -> VP PP

NP -> NP PP

NP -> NP and NP

PP-> P NPNP->Ahmad

NP->Lahore

NP->Karachi

NP->FebruaryVerb->flew

P->in

P->to

CONJ -> and